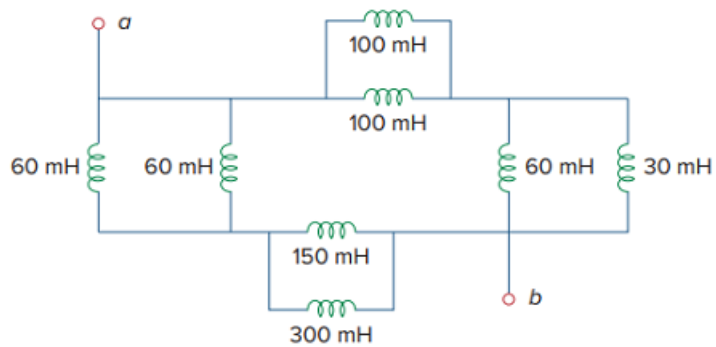


Problem

Find the equivalent inductance looking into the terminals of the circuit in Fig. 6.76.

Fig. 6.76:



Step-by-step solution

Step 1 of 3

Refer to Figure 6.76 in the textbook.

Consider 60 mH and 60 mH are in parallel combination.

Determine the equivalent inductance of 60 mH and 60 mH.

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

$$L_{eq1} = \frac{60 \times 60}{60 + 60}$$

$$= 30$$

Consider 100 mH and 100 mH are in parallel combination.

Determine the equivalent inductance of 100 mH and 100 mH.

$$L_{eq2} = \frac{100 \times 100}{100 + 100}$$

$$= 50$$

Consider 60 mH and 30 mH are in parallel combination.

Determine the equivalent inductance of 60 mH and 30 mH.

$$L_{eq3} = \frac{60 \times 30}{60 + 30}$$

$$= 20$$

Consider 150 mH and 300 mH are in parallel combination.

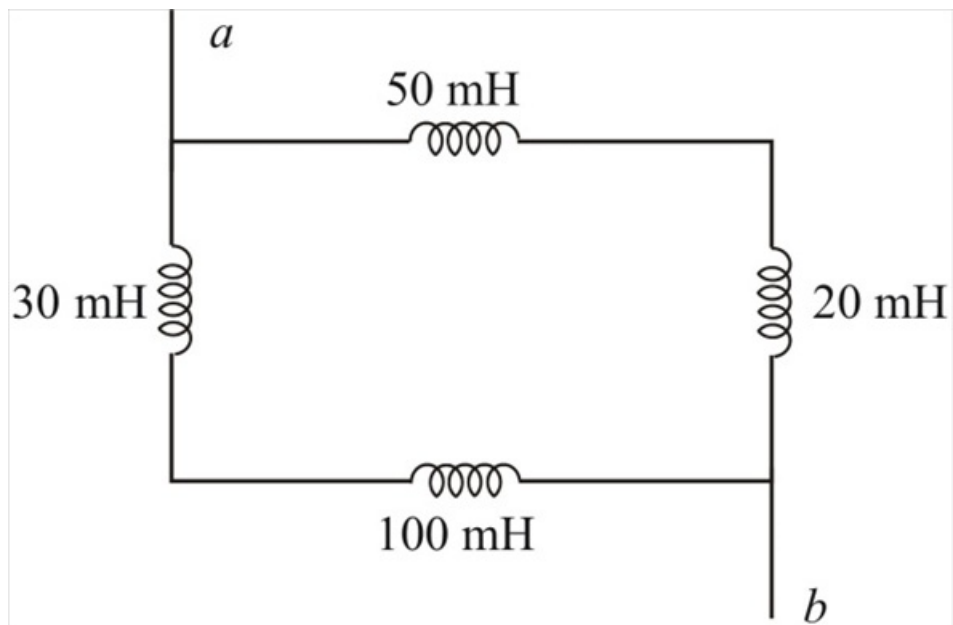
Determine the equivalent inductance of 150 mH and 300 mH.

$$L_{eq4} = \frac{150 \times 300}{150 + 300}$$

$$= 100$$

Step 2 of 3

Draw the circuit as shown in Figure.



Consider 50 mH and 20 mH are in series combination.

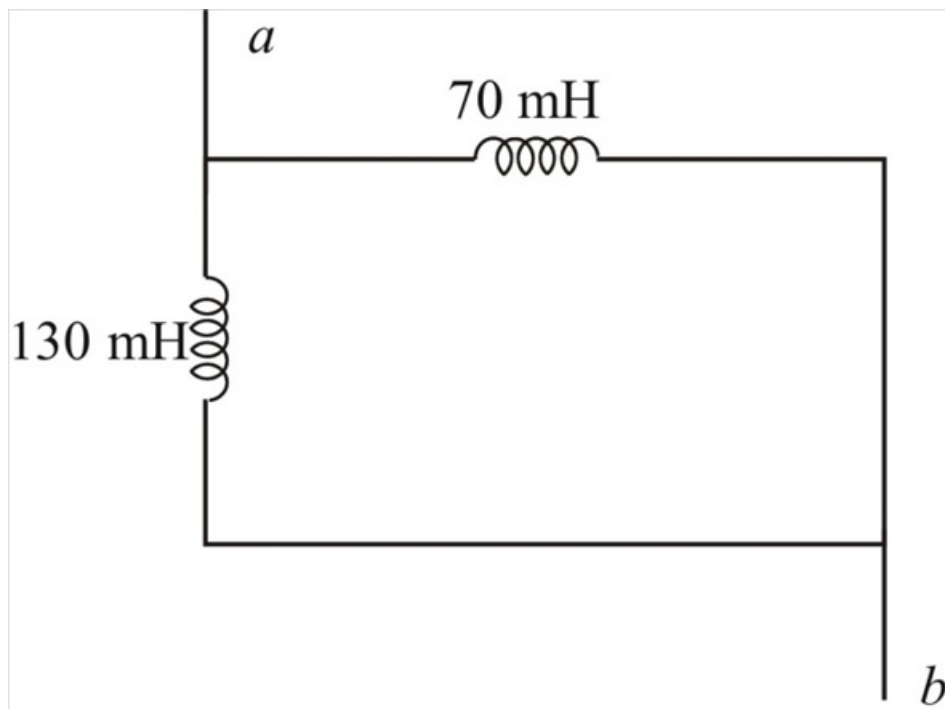
$$\begin{aligned}L_{eq5} &= L_1 + L_2 \\&= 50 + 20 \\&= 70 \text{ mH}\end{aligned}$$

Consider 30 mH and 100 mH are in series combination.

$$\begin{aligned}L_{eq5} &= L_1 + L_2 \\&= 30 + 100 \\&= 130 \text{ mH}\end{aligned}$$

Step 3 of 3

Draw the equivalent circuit as shown in Figure.



Finally 70 mH, 130 mH are in parallel

$$L_{eq} = \frac{70\text{m} \times 130\text{m}}{70\text{m} + 130\text{m}}$$

$$L_{eq} = 45.5 \text{ mH}$$

So, equivalent inductance of the given circuit is 45.5 mH.